

4.6.2 Polynomials and rational functions

Polynomial functions

A **polynomial function** is a function of the form

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

where $a_0, a_1, \dots, a_n \in \mathbb{R}$ are constants and $a_n \neq 0$. The number n is called the **degree** of the polynomial, denoted $\deg(p) = n$. The domain is $\text{dom}(p) = \mathbb{R}$.

- Example 4.7:**
1. Constant functions: $p(x) = c$ (degree 0).
 2. Linear functions: $p(x) = ax + b$ with $a \neq 0$ (degree 1).
 3. Quadratic functions: $p(x) = ax^2 + bx + c$ with $a \neq 0$ (degree 2).
 4. Cubic functions: $p(x) = ax^3 + bx^2 + cx + d$ with $a \neq 0$ (degree 3).

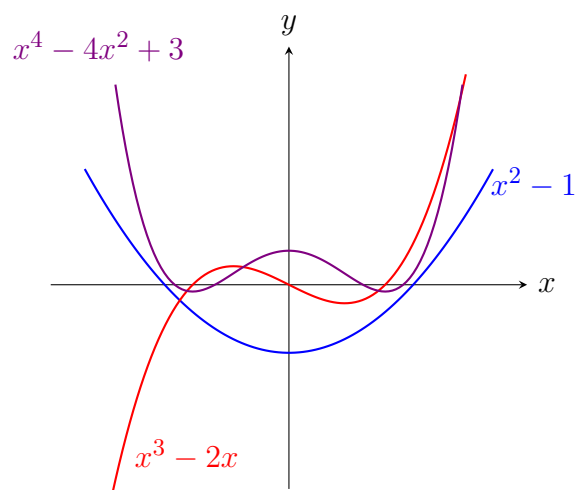


Figure 4.13: Examples of polynomial functions

Properties of polynomials

- The sum and product of two polynomials is a polynomial.
- If p has degree n , then p has at most n real roots (zeros).

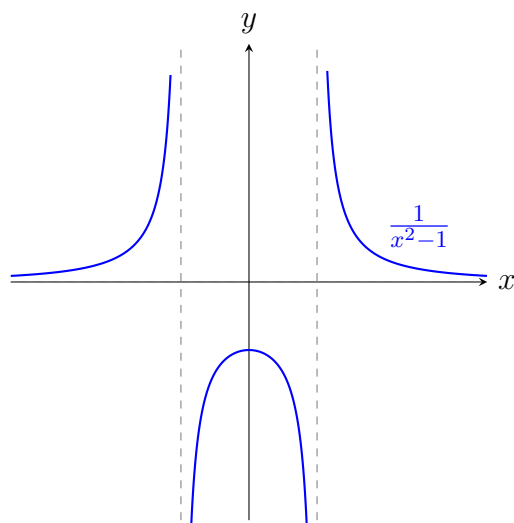
Rational functions

A **rational function** is a quotient of two polynomials:

$$r(x) = \frac{p(x)}{q(x)}$$

where p and q are polynomials with $q \not\equiv 0$. If p and q have no common factors then the domain is

$$\text{dom}(r) = \{x \in \mathbb{R} \mid q(x) \neq 0\}.$$



4.6.3 Exponential and logarithmic functions

The exponential function

For $a > 0$, $a \neq 1$, the **exponential function with base a** is

$$f(x) = a^x$$

with $\text{dom}(f) = \mathbb{R}$ and $\text{im}(f) = (0, +\infty)$. The most important case is $a = e$, where $e \approx 2.71828\dots$ is Euler's number. We write

$$\exp(x) = e^x.$$

Properties of exponential functions

For $a > 0$, $a \neq 1$:

- a^x is strictly increasing if $a > 1$ and strictly decreasing if $0 < a < 1$.
- $a^0 = 1$ for all $a > 0$.
- $a^x > 0$ for all $x \in \mathbb{R}$.

Algebraic rules for exponentials

For $a, b > 0$ and $x, y \in \mathbb{R}$:

$$a^x \cdot a^y = a^{x+y}$$

$$\frac{a^x}{a^y} = a^{x-y}$$

$$(a^x)^y = a^{xy}$$

$$a^x b^x = (ab)^x$$

$$a^{-x} = \frac{1}{a^x}$$

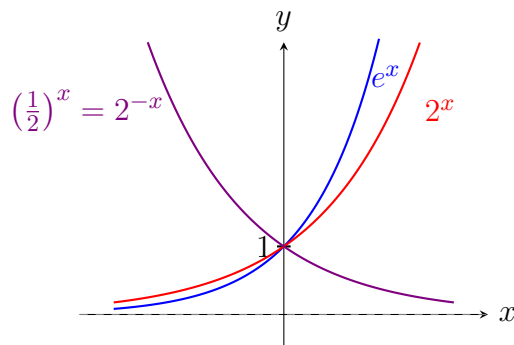


Figure 4.14: Exponential functions with different bases

The logarithmic function

For $a > 0$, $a \neq 1$, the **logarithm with base a** is the inverse of a^x :

$$f(x) = \log_a(x)$$

with $\text{dom}(f) = (0, +\infty)$ and $\text{im}(f) = \mathbb{R}$. The most important case is $a = e$, called the **natural logarithm**:

$$\ln(x) = \log_e(x).$$

Properties of logarithms

For $a > 0$, $a \neq 1$:

- $\log_a(x)$ is strictly increasing if $a > 1$ and strictly decreasing if $0 < a < 1$.
- $\log_a(1) = 0$ and $\log_a(a) = 1$ for all $a > 0$, $a \neq 1$.
- $\log_a(a^x) = x$ for all $x \in \mathbb{R}$ and $a^{\log_a(x)} = x$ for all $x > 0$.

Algebraic rules for logarithms

For $a > 0$, $a \neq 1$, and $x, y > 0$:

$$\log_a(xy) = \log_a(x) + \log_a(y)$$

$$\log_a\left(\frac{x}{y}\right) = \log_a(x) - \log_a(y)$$

$$\log_a(x^r) = r \log_a(x) \quad \text{for } r \in \mathbb{R}$$

$$\log_a(x) = \frac{\log_b(x)}{\log_b(a)} \quad (\text{change of base, } b > 0, b \neq 1)$$

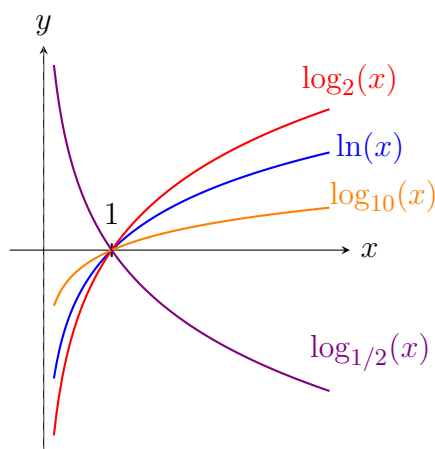


Figure 4.15: Logarithmic functions with different bases

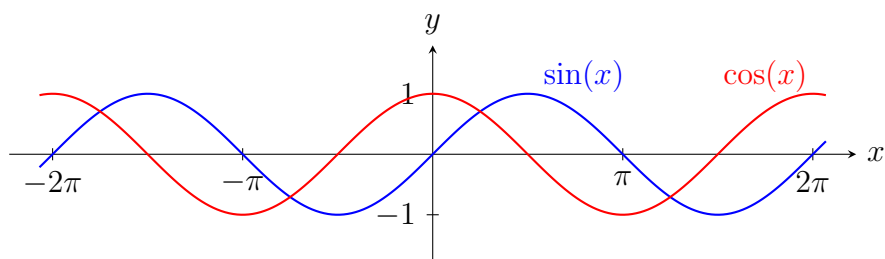
4.6.4 Trigonometric functions and their inverses

Sine and cosine

The **sine** and **cosine** functions are defined as

$$f(x) = \sin(x), \quad g(x) = \cos(x)$$

with $\text{dom}(f) = \text{dom}(g) = \mathbb{R}$ and $\text{im}(f) = \text{im}(g) = [-1, 1]$. Both functions are periodic with period 2π .



Other trigonometric functions

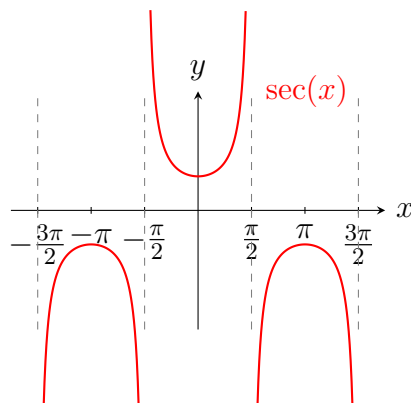
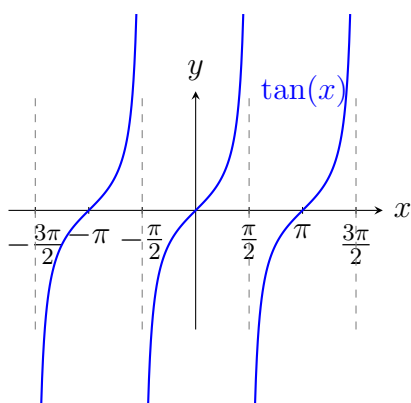
The other trigonometric functions are defined in terms of sine and cosine:

$$\tan(x) = \frac{\sin(x)}{\cos(x)} \quad \text{dom}(\tan) = \mathbb{R} \setminus \left\{ \frac{\pi}{2} + k\pi \mid k \in \mathbb{Z} \right\}$$

$$\cot(x) = \frac{\cos(x)}{\sin(x)} \quad \text{dom}(\cot) = \mathbb{R} \setminus \{k\pi \mid k \in \mathbb{Z}\}$$

$$\sec(x) = \frac{1}{\cos(x)} \quad \text{dom}(\sec) = \mathbb{R} \setminus \left\{ \frac{\pi}{2} + k\pi \mid k \in \mathbb{Z} \right\}$$

$$\csc(x) = \frac{1}{\sin(x)} \quad \text{dom}(\csc) = \mathbb{R} \setminus \{k\pi \mid k \in \mathbb{Z}\}$$



Fundamental trigonometric identities

$$\sin^2(\alpha) + \cos^2(\alpha) = 1$$

$$\tan^2(\alpha) + 1 = \sec^2(\alpha)$$

$$1 + \cot^2(\alpha) = \csc^2(\alpha)$$

$$\sin(\alpha \pm \beta) = \sin(\alpha) \cos(\beta) \pm \cos(\alpha) \sin(\beta)$$

$$\cos(\alpha \pm \beta) = \cos(\alpha) \cos(\beta) \mp \sin(\alpha) \sin(\beta)$$

$$\sin(2\alpha) = 2 \sin(\alpha) \cos(\alpha)$$

$$\cos(2\alpha) = \cos^2(\alpha) - \sin^2(\alpha) = 2 \cos^2(\alpha) - 1 = 1 - 2 \sin^2(\alpha)$$

$$\sin(-\alpha) = -\sin(\alpha) \quad (\text{sine is odd})$$

$$\cos(-\alpha) = \cos(\alpha) \quad (\text{cosine is even})$$

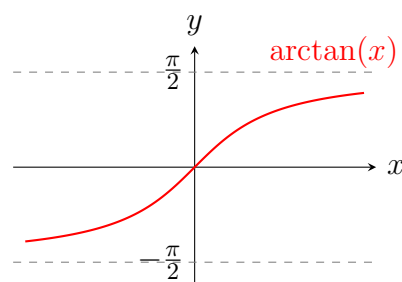
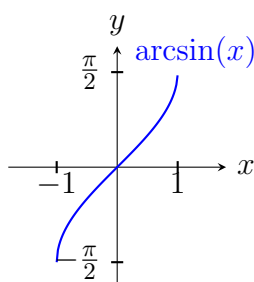
$$\sin(\alpha) - \sin(\beta) = 2 \cos\left(\frac{\alpha + \beta}{2}\right) \sin\left(\frac{\alpha - \beta}{2}\right)$$

$$\cos(\alpha) - \cos(\beta) = -2 \sin\left(\frac{\alpha + \beta}{2}\right) \sin\left(\frac{\alpha - \beta}{2}\right)$$

Inverse trigonometric functions

The inverse trigonometric functions are defined by restricting the domains appropriately:

- $\arcsin : [-1, 1] \rightarrow [-\frac{\pi}{2}, \frac{\pi}{2}]$ is the inverse of $\sin$ restricted to $[-\frac{\pi}{2}, \frac{\pi}{2}]$.
- $\arccos : [-1, 1] \rightarrow [0, \pi]$ is the inverse of $\cos$ restricted to $[0, \pi]$.
- $\arctan : \mathbb{R} \rightarrow (-\frac{\pi}{2}, \frac{\pi}{2})$ is the inverse of $\tan$ restricted to $(-\frac{\pi}{2}, \frac{\pi}{2})$.



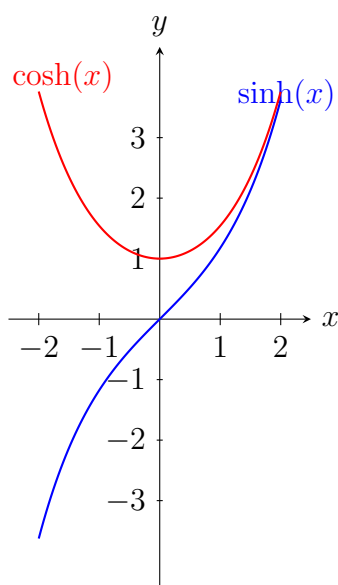
4.6.5 Hyperbolic functions

Hyperbolic sine and cosine

The **hyperbolic sine** and **hyperbolic cosine** functions are defined as

$$f(x) = \sinh(x) = \frac{e^x - e^{-x}}{2}, \quad g(x) = \cosh(x) = \frac{e^x + e^{-x}}{2}$$

with $\text{dom}(f) = \text{dom}(g) = \mathbb{R}$, $\text{im}(f) = \mathbb{R}$ and $\text{im}(g) = [1, +\infty)$. Neither function is periodic.



Other hyperbolic functions

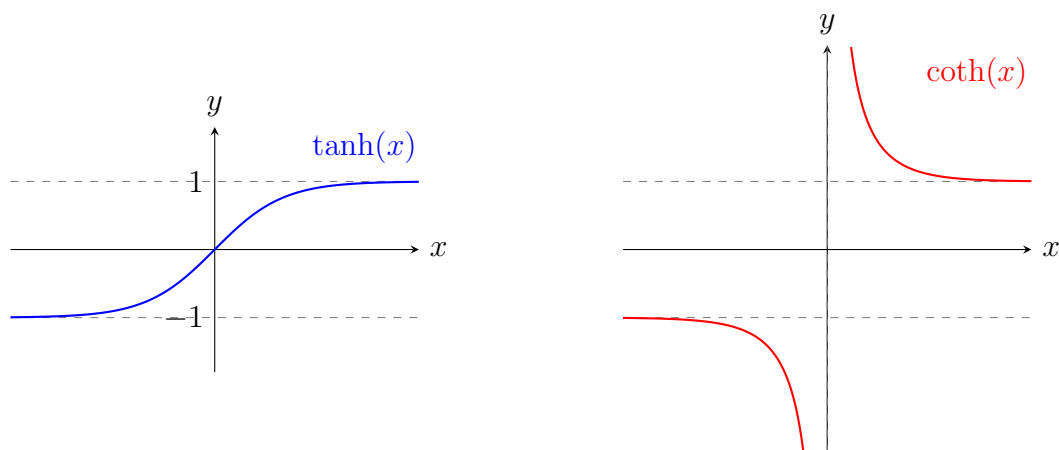
The other hyperbolic functions are defined in terms of hyperbolic sine and cosine:

$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)} \quad \text{dom}(\tanh) = \mathbb{R}$$

$$\coth(x) = \frac{\cosh(x)}{\sinh(x)} \quad \text{dom}(\coth) = \mathbb{R} \setminus \{0\}$$

$$\text{sech}(x) = \frac{1}{\cosh(x)} \quad \text{dom}(\text{sech}) = \mathbb{R}$$

$$\text{csch}(x) = \frac{1}{\sinh(x)} \quad \text{dom}(\text{csch}) = \mathbb{R} \setminus \{0\}$$



Fundamental hyperbolic identities

$$\cosh^2(\alpha) - \sinh^2(\alpha) = 1$$

$$1 - \tanh^2(\alpha) = \operatorname{sech}^2(\alpha)$$

$$\coth^2(\alpha) - 1 = \operatorname{csch}^2(\alpha)$$

$$\sinh(\alpha \pm \beta) = \sinh(\alpha) \cosh(\beta) \pm \cosh(\alpha) \sinh(\beta)$$

$$\cosh(\alpha \pm \beta) = \cosh(\alpha) \cosh(\beta) \pm \sinh(\alpha) \sinh(\beta)$$

$$\sinh(2\alpha) = 2 \sinh(\alpha) \cosh(\alpha)$$

$$\cosh(2\alpha) = \cosh^2(\alpha) + \sinh^2(\alpha) = 2 \cosh^2(\alpha) - 1 = 1 + 2 \sinh^2(\alpha)$$

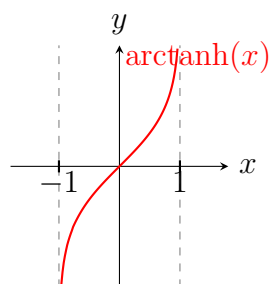
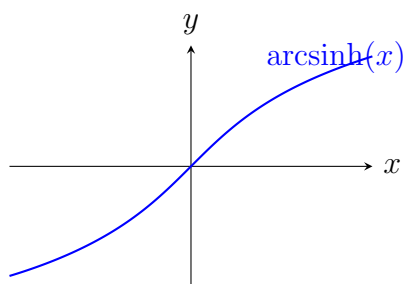
$$\sinh(-\alpha) = -\sinh(\alpha) \quad (\text{hyperbolic sine is odd})$$

$$\cosh(-\alpha) = \cosh(\alpha) \quad (\text{hyperbolic cosine is even})$$

Inverse hyperbolic functions

The inverse hyperbolic functions are defined by restricting the domains appropriately, and can be expressed in terms of logarithms:

- $\operatorname{arcsinh} : \mathbb{R} \rightarrow \mathbb{R}$ is the inverse of $\sinh$, given by $\operatorname{arcsinh}(x) = \ln \left(x + \sqrt{x^2 + 1} \right)$.
- $\operatorname{arccosh} : [1, +\infty) \rightarrow [0, +\infty)$ is the inverse of $\cosh$ restricted to $[0, +\infty)$, given by $\operatorname{arccosh}(x) = \ln \left(x + \sqrt{x^2 - 1} \right)$.
- $\operatorname{arctanh} : (-1, 1) \rightarrow \mathbb{R}$ is the inverse of $\tanh$, given by $\operatorname{arctanh}(x) = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$.



Chapter 5

Limits and continuity

The concepts of limits and continuity lie at the very foundation of mathematical analysis. They formalize the intuitive ideas of *approaching* and *connectedness*, and provide the rigorous language needed to define derivatives and integrals.

Historically, these ideas evolved over centuries. Ancient Greek mathematicians like Eudoxus and Archimedes used rudimentary forms of limits in the *method of exhaustion* to compute areas and volumes. However, it was not until the 17th century, with the development of calculus by Newton and Leibniz, that the notion of a limit became central—though it remained informal and often controversial.

In the 19th century, mathematicians such as Cauchy and Weierstrass finally placed limits on a firm logical foundation using the famous ε - δ definitions. The key insight was to quantify the idea of “arbitrarily close” using two positive numbers: ε for the error tolerance in the function’s output, and δ for the corresponding distance in the input. This rigorous framework eliminated the paradoxes and ambiguities that had plagued early calculus and remains the standard today.

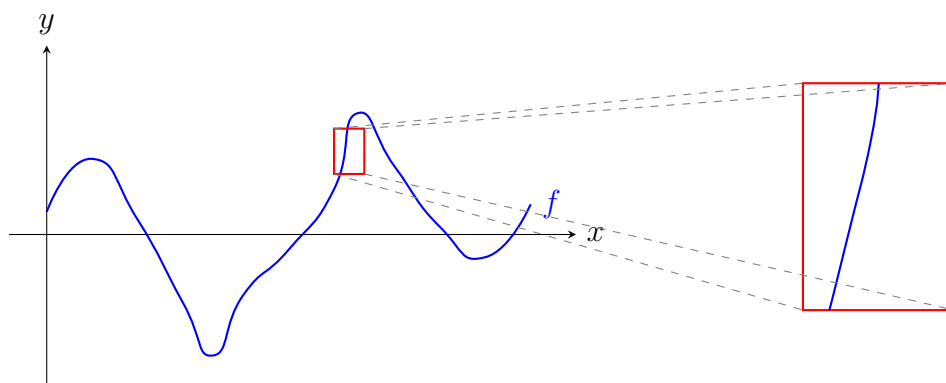


Figure 5.1: We will develop rigorous tools for “zooming-in”

5.1 Neighborhoods

To discuss continuity and limits, we need to be able to “zoom” into the graph of a function. To do this, we need a definition of a small subset around a given point that we want to investigate. This is called a *neighborhood*: